MINI PROJECT 3

Northwest Architecture Studio Network

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Introduction:

This project focuses on designing and implementing a refreshed, modern computer network for NorthWest Architecture Studio Ltd. The studio employs a diverse team of professionals, including administrative staff, project managers, CAD/designers, and 3D visualization artists. To keep business operations running smoothly, these employees

must be able to securely share huge design files with one another inside the office while safely accessing external cloud services on the internet.

To model this environment, our team used the Filius network simulation program. This report documents how we built the network, configured its software settings, methodically tested its connections, and verified that data can travel seamlessly across different network borders.

Objectives:

Three main goals to achieve during this project:

- **Network Separation:** Keep the internal office files hidden away from the open internet for safety.
- **Service Availability:** Set up a local internal website (an intranet) and a working network phonebook (DNS server) to turn text website names into network numbers.
- **Cross-Network Communication:** Prove that data can travel through a router from an internal workstation to a simulated public cloud server without needing overly complex configurations.
- **Client-Server Functionality:** Configure servers to provide services to client devices and demonstrate how requests and responses are handled across the network.
- **Network Testing:** Verify network connectivity through ping tests, web browser access, and DNS name resolution.

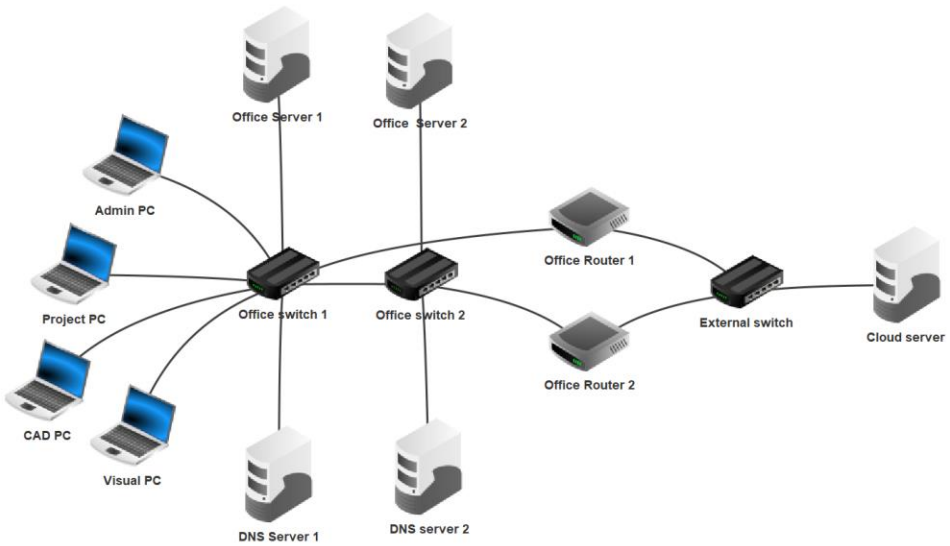
Network Topology:

The network consists of:

- Office Router
- Office router 2 (backup)
- Office Switch
- Office Switch 2 (backup)
- External / ISP Switch
- Admin PC
- Project PC
- CAD PC
- Visual PC
- DNS Server
- DNS Server 2 (backup)
- Office Server
- Office Server 2 (backup)
- ISP / Cloud Server

The Office Router connects the internal LAN to the external cloud network. Office devices communicate through the Office Switch, while the cloud server is connected through the External / ISP Switch.


90%


Office Server 1
Office Server 2
Admin PC
Office Router 1
Project PC
Office switch 1
Office switch 2
External switch
Cloud server
Office Router 2
CAD PC
Visual PC
DNS Server 1
DNS server 2

IP Addressing Table:

The network consists of an Office Router, two switches, four client computers, a DNS server, an Office Server, and an ISP/Cloud Server. The internal Office LAN uses the 192.168.10.0/24 network range, while the external cloud network uses the 203.0.113.0/24 network range. The router acts as the gateway between the two networks and allows data to travel between them. DNS services are used to convert domain names into IP addresses, making resources easier for users to access.

Device	IP Address	Subnet Mask	Gateway	DNS
Router Interface 1	192.168.10.1	255.255.255.0	-	-
Router Interface 2	203.0.113.1	255.255.255.0	-	-
DNS Server	192.168.10.2	255.255.255.0	192.168.10.1	192.168.10.2
Office Server	192.168.10.10	255.255.255.0	192.168.10.1	192.168.10.2
Admin PC	192.168.10.101	255.255.255.0	192.168.10.1	192.168.10.2
Project PC	192.168.10.111	255.255.255.0	192.168.10.1	192.168.10.2
CAD PC	192.168.10.121	255.255.255.0	192.168.10.1	192.168.10.2
Visual PC	192.168.10.131	255.255.255.0	192.168.10.1	192.168.10.2
ISP / Cloud Server	203.0.113.10	255.255.255.0	203.0.113.1	192.168.10.2

This addressing scheme uses two separate /24 networks. The office network uses 192.168.10.0/24 while the cloud network uses 203.0.113.0/24. Communication between these networks is achieved through the router.

DNS Configurations:

The DNS (Domain Name System) server was configured to provide name resolution for both internal and external services. Instead of requiring users to remember IP addresses, DNS allows resources to be accessed using meaningful domain names. The record **intranet.archstudio.local** points to the internal Office Server, while **cloud.archstudio.example** points to the external Cloud Server. This improves usability and reflects how DNS functions in modern business networks by translating domain names into IP addresses.

DNS Name	IP Address
intranet.archstudio.local	192.168.10.10

cloud.archstudio.example	203.0.113.10

These records allow users to access services using human-readable names instead of IP addresses.

Client Server Architect:

The network follows a client-server architecture, where client devices request services from dedicated servers. In this simulation, the Admin PC, Project PC, CAD PC and Visual PC act as client devices, while the DNS Server, Office Server and Cloud Server provide services. The Office Server hosts the company's internal intranet website, while the Cloud Server provides access to external resources. When a user accesses a website, the client sends a request to the server, which processes the request and returns the required information. This model demonstrates how organizations centralize resources and services for efficient management, security, and accessibility.

Examples:

- CAD PC requests web pages from Office Server.
- CAD PC requests cloud services from ISP/Cloud Server.
- Client devices request name resolution from DNS Server.

The servers provide resources while the clients consume those resources, demonstrating the client-server architecture used in modern networks.

Protocols Explanations:

A protocol is a standardized set of rules that allows different devices and software programs to communicate with each other; all devices need to follow the same set of rules. If they don't follow the same rules, they cannot talk to each other. For example, in our Filius network, the CAD PC used DNS to find the IP address of the office server, then used HTTP to request the intranet page from the office server because that is where the page is stored.

Concept	What is it?	How it was used in our build
TCP/IP	TCP/IP Stands for transmission control Protocol / Internet Protocol. It is a foundational set of communication rules that allows computers and devices to connect and share data to communicate on the internet. IP provides the address system and routing. It gives a unique address to devices And TCP handles the delivery of the data, to make sure the packets arrive safely.	Every device had a unique IP address, for example the CAD PC was 192.168.10.121. When it opened the cloud page, IP sent packets through the office to reach 203.0.113.10 And the TCP made sure the page arrived complete.
IP addresses	An IP address is a unique number given to every device on the network. When a device wants to send data, it needs the IP address of the other device, like a postal address on a letter. Routers use IP addresses to find the right path to the destination. An IP address has two parts: the network part (192.168.10) And the device part (.121).	In our Filius network, every device has a unique IP address. The office devices were on the 192.168.10.0 network, for example the CAD PC was 192.168.10.121. And the office server was 192.168.10.10. The cloud Server was on a different network: 203.0.113.10.
DNS	DNS turns names into numbers (IP addresses), because humans prefer to use names, but computers only understand numbers.	Our DNS Servers were DNS Server 1 (192.168.10.2) and DNS Server 2 (192.168.10.3). Each one had two records intranet.archstudio.example

		(192.168.10.10) and cloud.archstudio.example (203.0.113.10). When the CAD PC typed a name in the browser, it asked the DNS Server for the IP address first.
HTTP	HTTP is the protocol a web browser uses to request a web page from a server. In this case the servers are the Office Server and the Cloud server	The CAD PC used HTTP to open two pages. First the DNS found the IP address of each name, then the browser sent an HTTP request. The page at (HTTP:// intranet.archstudio.local Was sent by the Office Server (192.168.10.10), and the page at https;//cloud.archstudio.example was sent by the Cloud server (203.0.113.10). Each page is an index.html file stored on the server's webserver.
ICMP	Ping is used to send a signal over the network to see if a device can be reached. It uses the ICMP (Internet Control Message Protocol). TTL means time to live, and it controls how long the data should stay live before it expires. Every time the data makes a hop through a router, the TTL goes down by 1.	We pinged the office server, the DNS Server, the router, and the Cloud Server from the CAD PC, and they all replied with 0% loss. Inside the Office Network, the TTL was 64, but the cloud server was 63 because the data had to make one hop through the Office Router to reach the external network.
Gateway	The gateway is like an exit door to the outside. A PC with no Gateway can only talk to its own LAN. With a gateway, it can talk to other devices outside the network. And a router itself did not need a gateway because it is directly connected to both networks.	In our Filius network, every office device uses (192.168.10.1) the office router as its gateway, and the Cloud Server used 203.0.113.1. This is how the CAD PC could talk to the Cloud Server, which represents the internet in our network.
Switch	A switch is a hardware device that connects multiple computers, servers, and other devices inside a LAN, so they can communicate with each other. It uses MAC addresses	In our Filius network, we have three switches. Office Switch 1 connects to the four PCs, office Server 1 and DNS Server 1. Office Switch 2 is a backup for redundancy. If Switch 1 fails, the

	to send data only to the correct device.	backup Server (Office Server 2 and DNS Server 2) is still connected through Switch 2 and Router 2. The external Switch connects the routers to the Cloud Server.
Router	A router is a network device that helps devices connect to an external network, such as the internet. It sends data in packets using IP addresses.	In our Filius network, Office Router 1 has two IP addresses: 192.168.10.1 on the Office Switch 1 side and 203.0.113.1 on the external side, connected to the External Switch. This lets it join the Office LAN to the external network. We added Office Router 2 (192.168.10.254 and 203.0.113.254) as a backup for redundancy. If router 1 goes down, devices can change their gateways to router 2 and still reach the external network.

Testing Table:

Test	Command / Action	Expected Result	Actual Result	Outcome
LAN PC-to-PC	ping 192.168.10.101	Reply from Admin PC	Reply received successfully	Pass
PC to Office Server	ping 192.168.10.10	Reply from Office Server	Reply received successfully	Pass
PC to DNS Server	ping 192.168.10.2	Reply from DNS Server	Reply received successfully	Pass
External Server Ping	ping 203.0.113.10	Reply from Cloud Server	Reply received successfully	Pass

Internal Web by IP	http://192.168.10.10	Website loads	Website loaded successfully	Pass
Internal Web by Name	http://intranet.archstudio.local	Website loads through DNS	DNS resolved and page loaded	Pass
External Web by IP	http://203.0.113.10	Website loads	Website loaded successfully	Pass
External Web by Name	http://cloud.archstudio.example	Website loads through DNS	DNS resolved and page loaded	Pass

Screenshot (Evidence of the testing):

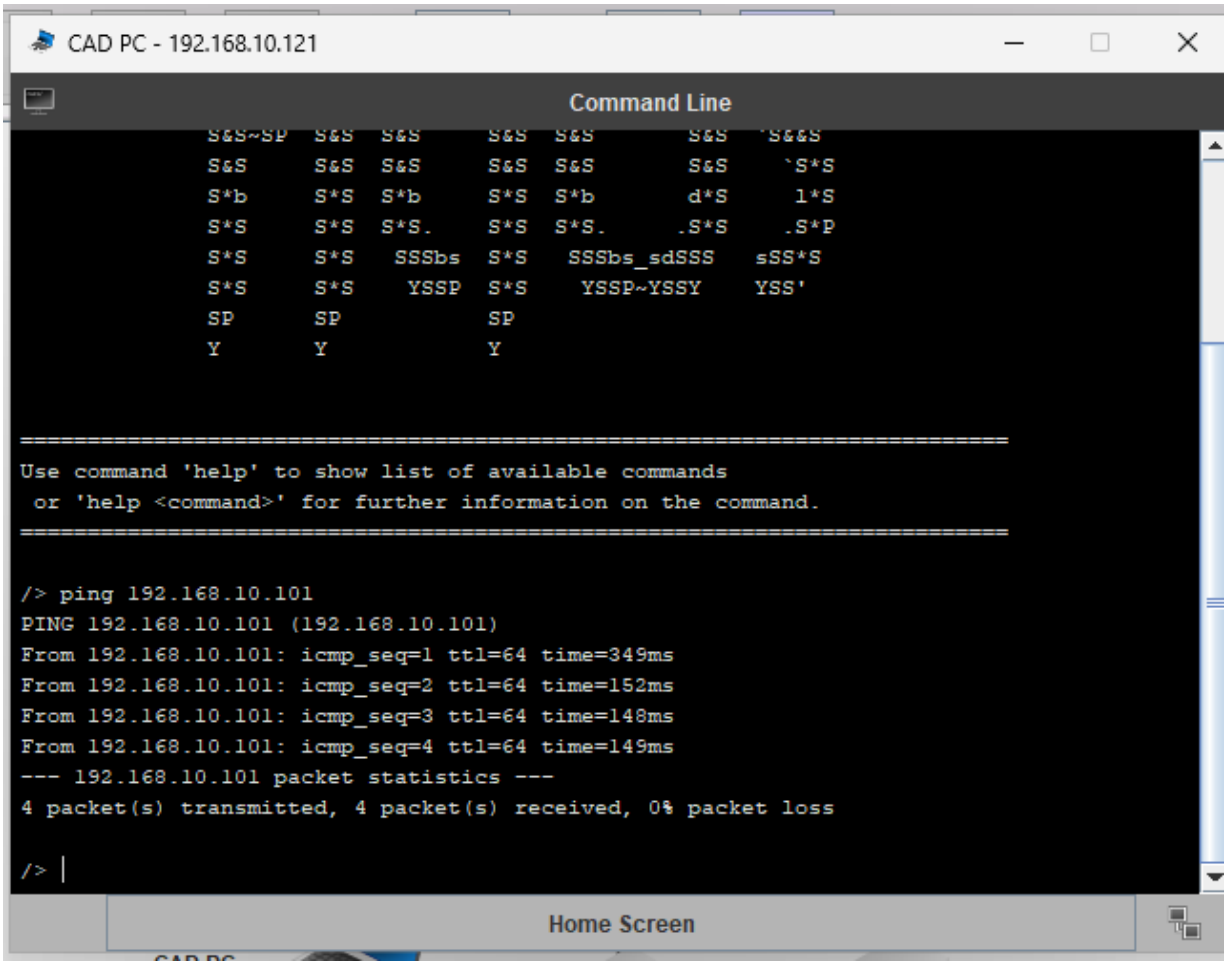


Figure 1: *LAN Ping test*

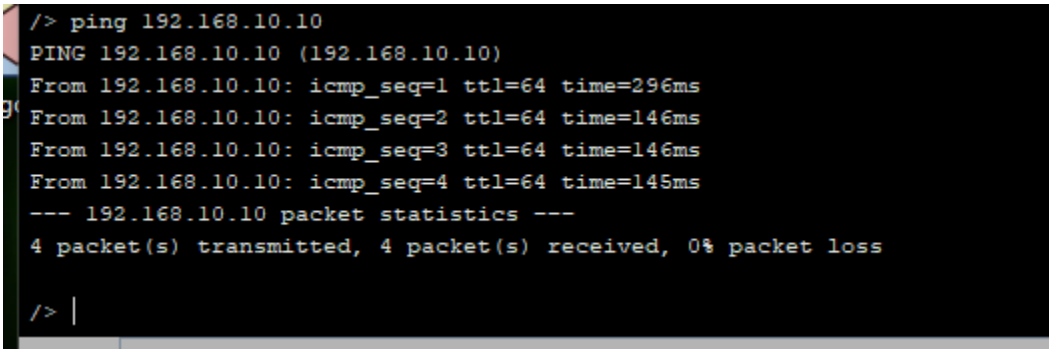


Figure 2: *Office Server Ping*

```
/> ping 192.168.10.2
PING 192.168.10.2 (192.168.10.2)
From 192.168.10.2: icmp_seq=1 ttl=64 time=300ms
From 192.168.10.2: icmp_seq=2 ttl=64 time=147ms
From 192.168.10.2: icmp_seq=3 ttl=64 time=149ms
From 192.168.10.2: icmp_seq=4 ttl=64 time=148ms
--- 192.168.10.2 packet statistics ---
4 packet(s) transmitted, 4 packet(s) received, 0% packet loss
```

Figure 3: DNS *Server Ping*

```
/> ping 203.0.113.10
PING 203.0.113.10 (203.0.113.10)
From 203.0.113.10: icmp_seq=1 ttl=63 time=592ms
From 203.0.113.10: icmp_seq=2 ttl=63 time=294ms
From 203.0.113.10: icmp_seq=3 ttl=63 time=297ms
From 203.0.113.10: icmp_seq=4 ttl=63 time=295ms
--- 203.0.113.10 packet statistics ---
4 packet(s) transmitted, 4 packet(s) received, 0% packet loss
```

Figure 4: *External Cloud Ping*

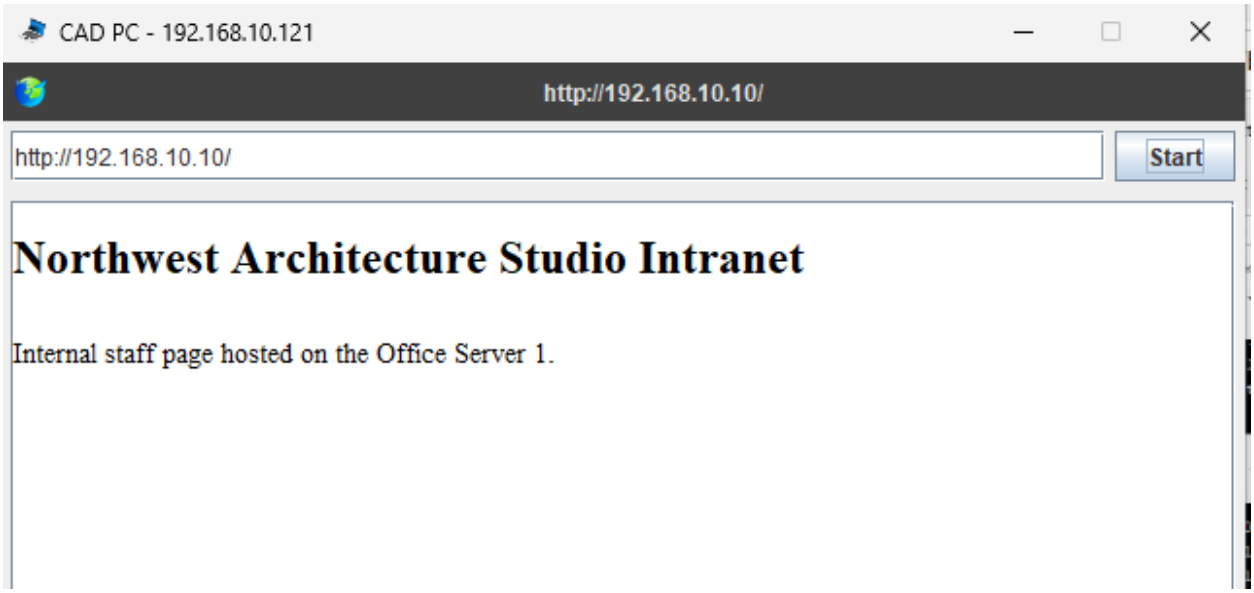


Figure 5: *Internal website by IP*

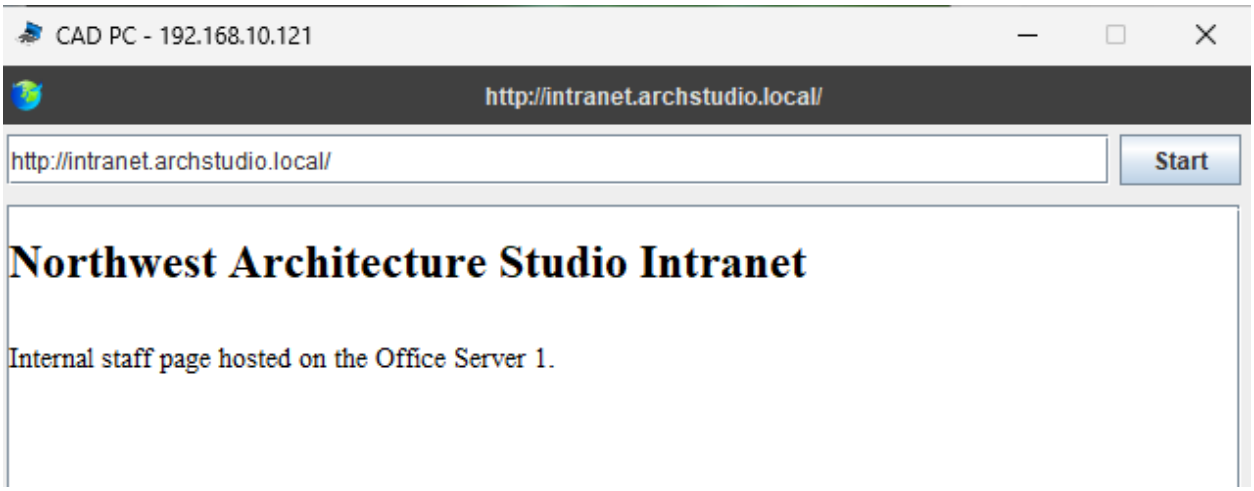


Figure 6: *Internal website by Name*

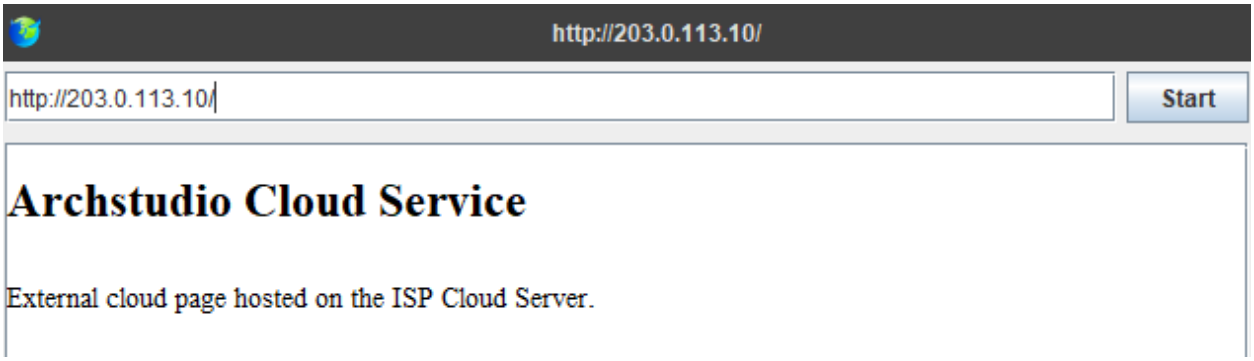


Figure 7: *External website by IP*



Figure 8: *External website by Name*

Troubleshooting log:

Issue Identified	Cause	Resolution	Result
Internal intranet page could not be accessed	DNS record contained a typo in the domain name	Corrected the DNS record for intranet.archstudio.local	Website loaded successfully
Backup office server was unavailable after primary server failure	DNS record still pointed to Office Server 1	Updated DNS record to Office Server 2 (192.168.10.11)	Intranet became accessible again
DNS name resolution failed during testing	Incorrect DNS entry configuration	Corrected DNS records on DNS Server	Domain names resolved successfully

Redundancy

Redundancy refers to having an extra, backup, or secondary system ready in case something goes wrong. Redundancy can improve local network reliability. By reducing dependence on a single device. But the backup server must contain an up-to-date copy of the files. And the network must be configured correctly for the backup service to work. But in this project, switching to the backup file server is performed manually.

For Northwest Architecture studio, if the office server fails, CAD designers cannot open their project files. We added 2 Switches, 2 Routers, 2 DNS servers, and 2 office servers.

Demo1: office Server failure

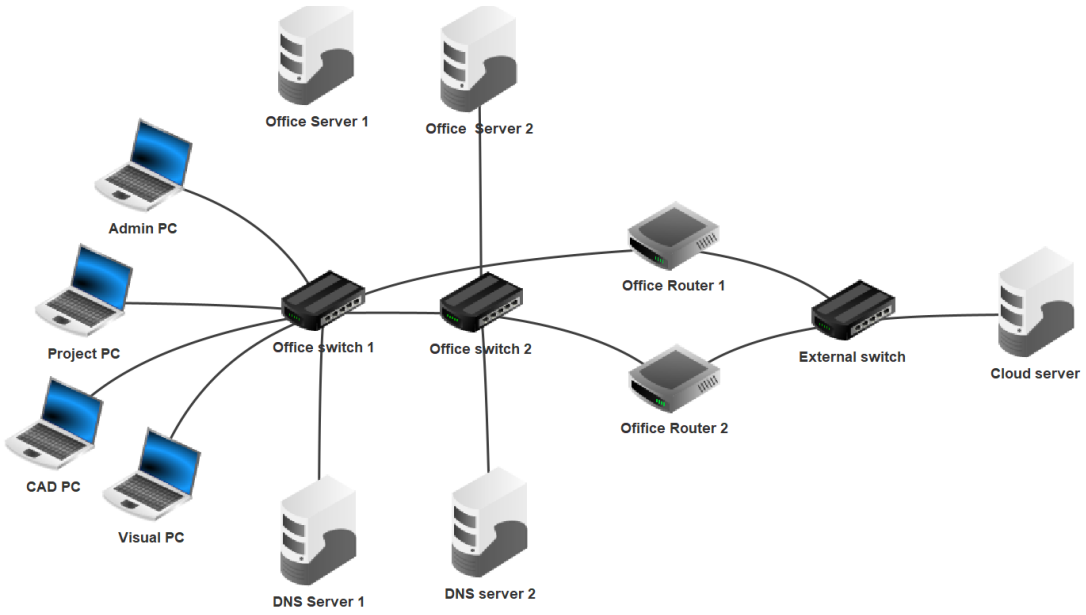


Figure 9: *office Server 1 Cable removed to simulate a Server failure*.

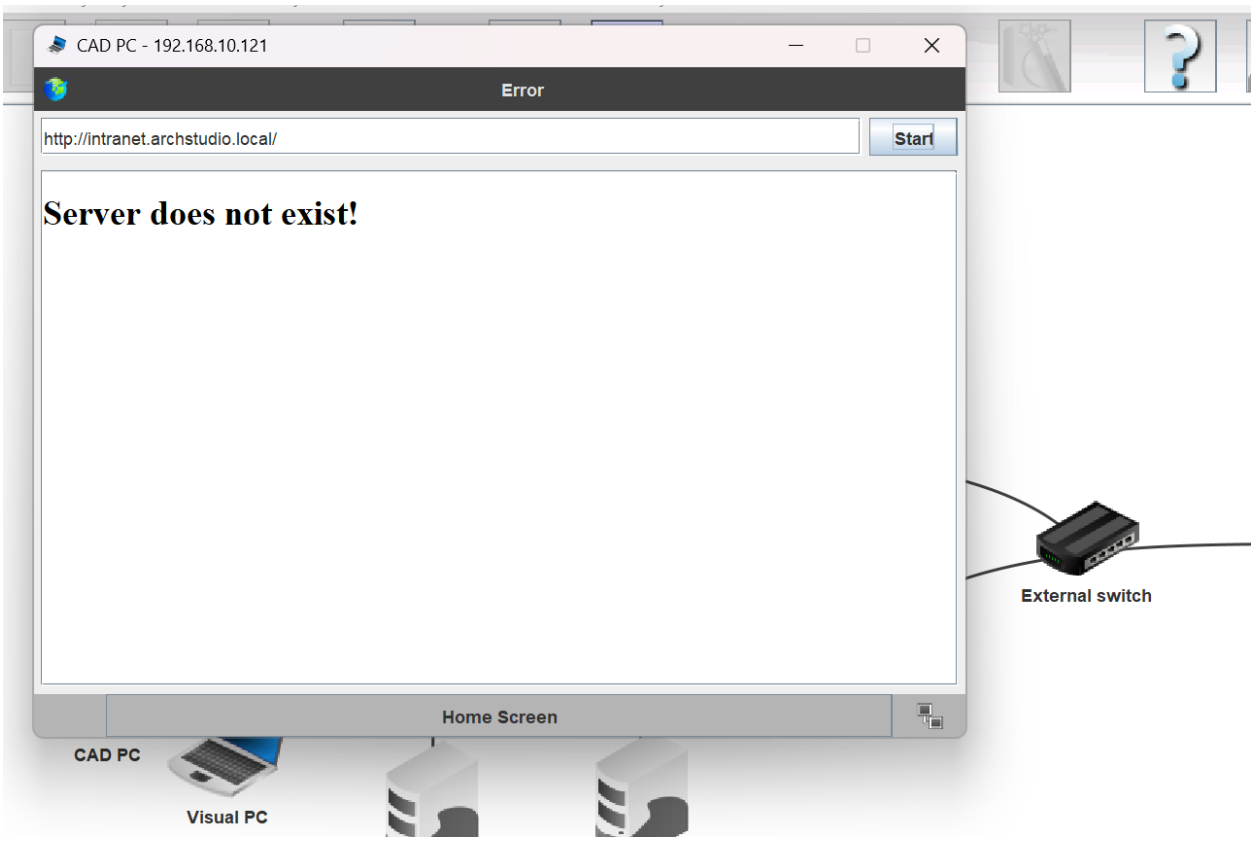


Figure 10: *The intranet cannot be reached "Server does not exist"*

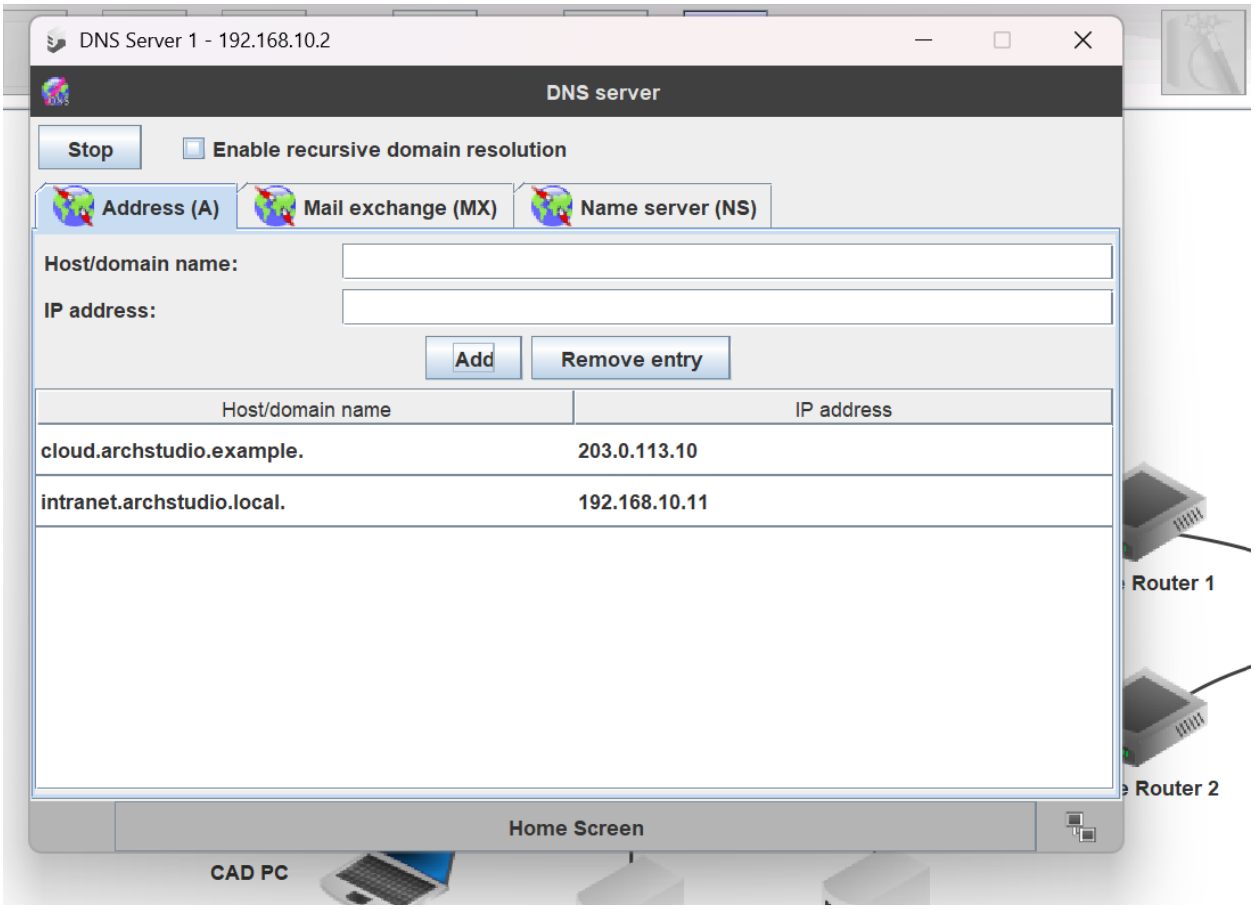


***Figure 11**: The DNS record for intranet changed to 192.168.10.11 (office server 2).*

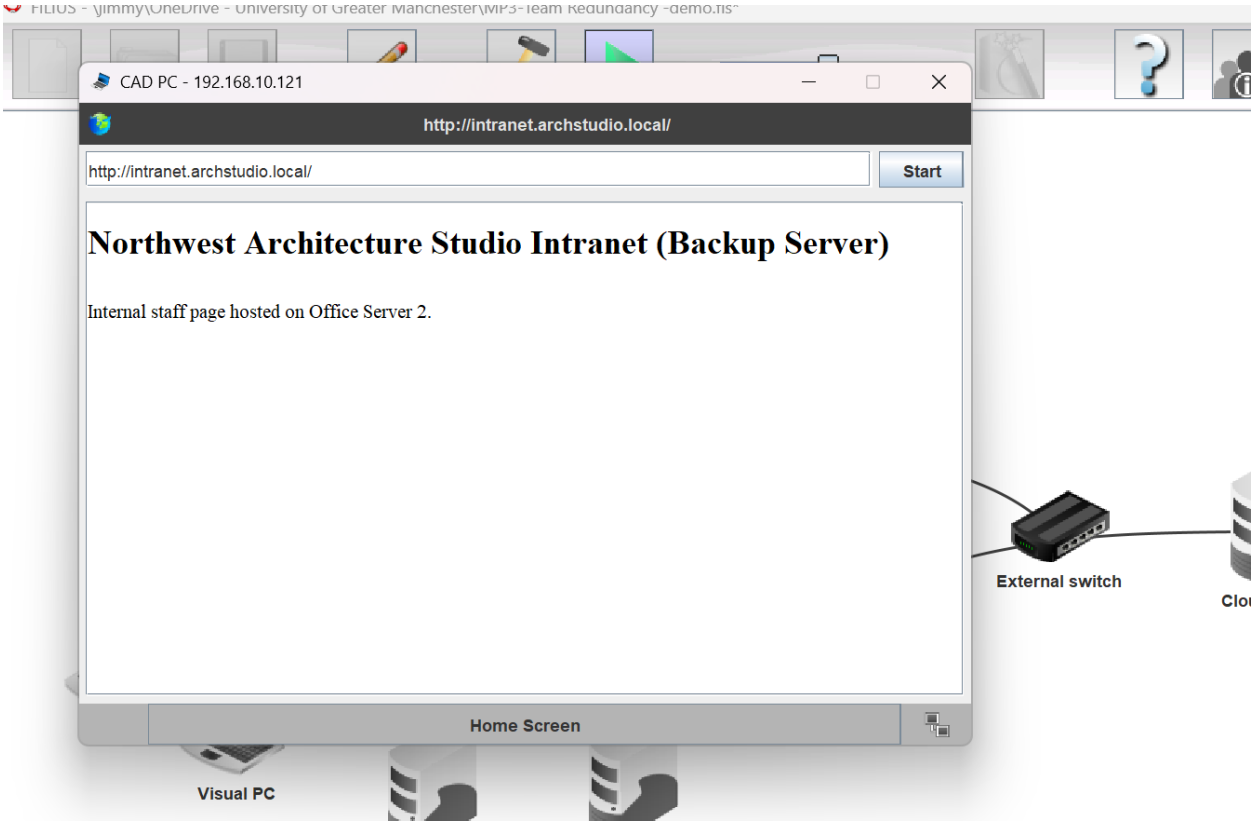


Figure 12: *After changing the DNS record to 192.168.10.11, the intranet loads from the backup Server (Office Server 2)*

Demo2: DNS Server failure

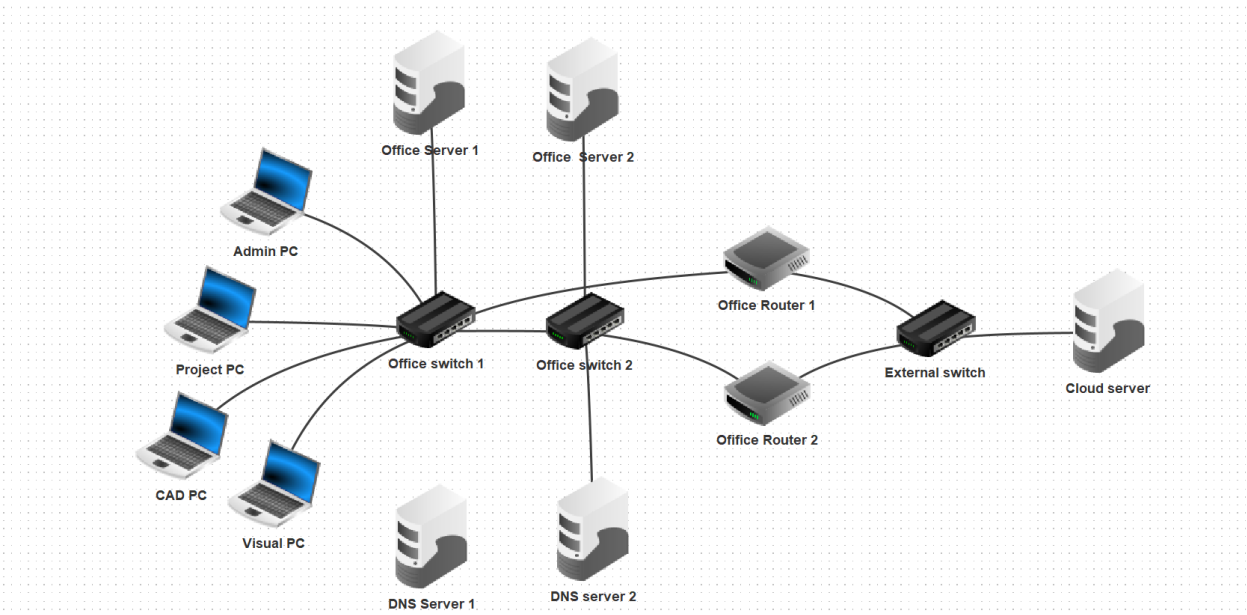


Figure 13: DNS Server 1 cable removed to Simulate a DNS Server failure

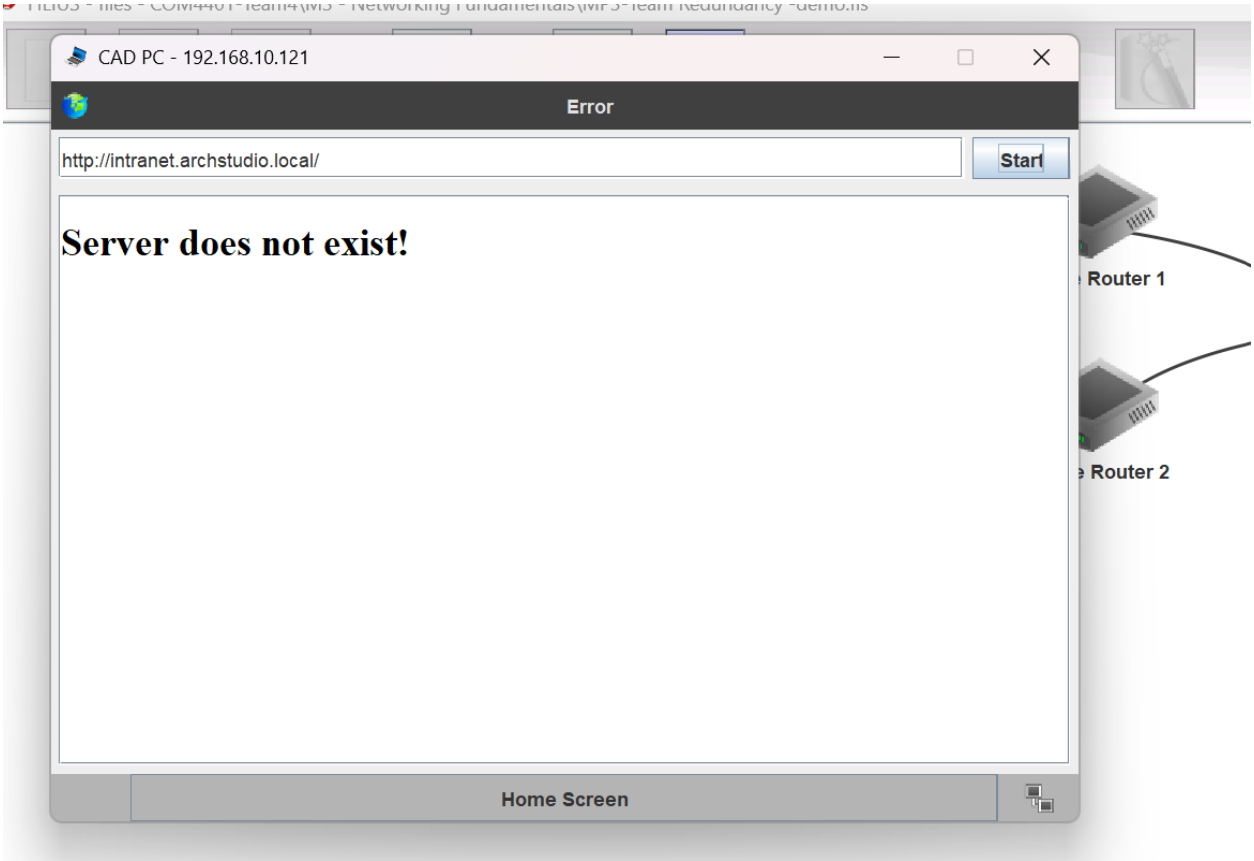


Figure 14: *With DNS Server 1 down, the CAD PC cannot open the intranet by name ("Server does not exist").*

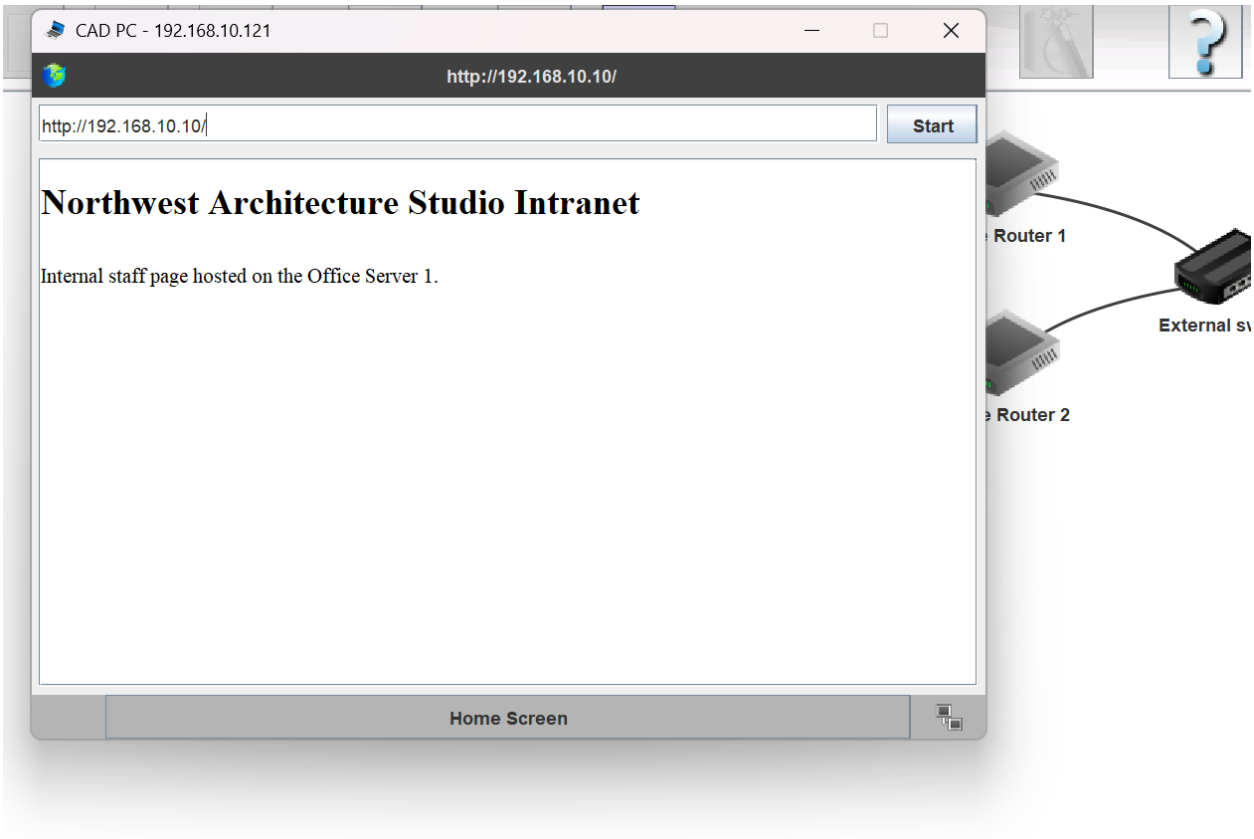


Figure 15: *The intranet IP address (192.168.10.10). This shows that the web Server is working, and only DNS has failed.*

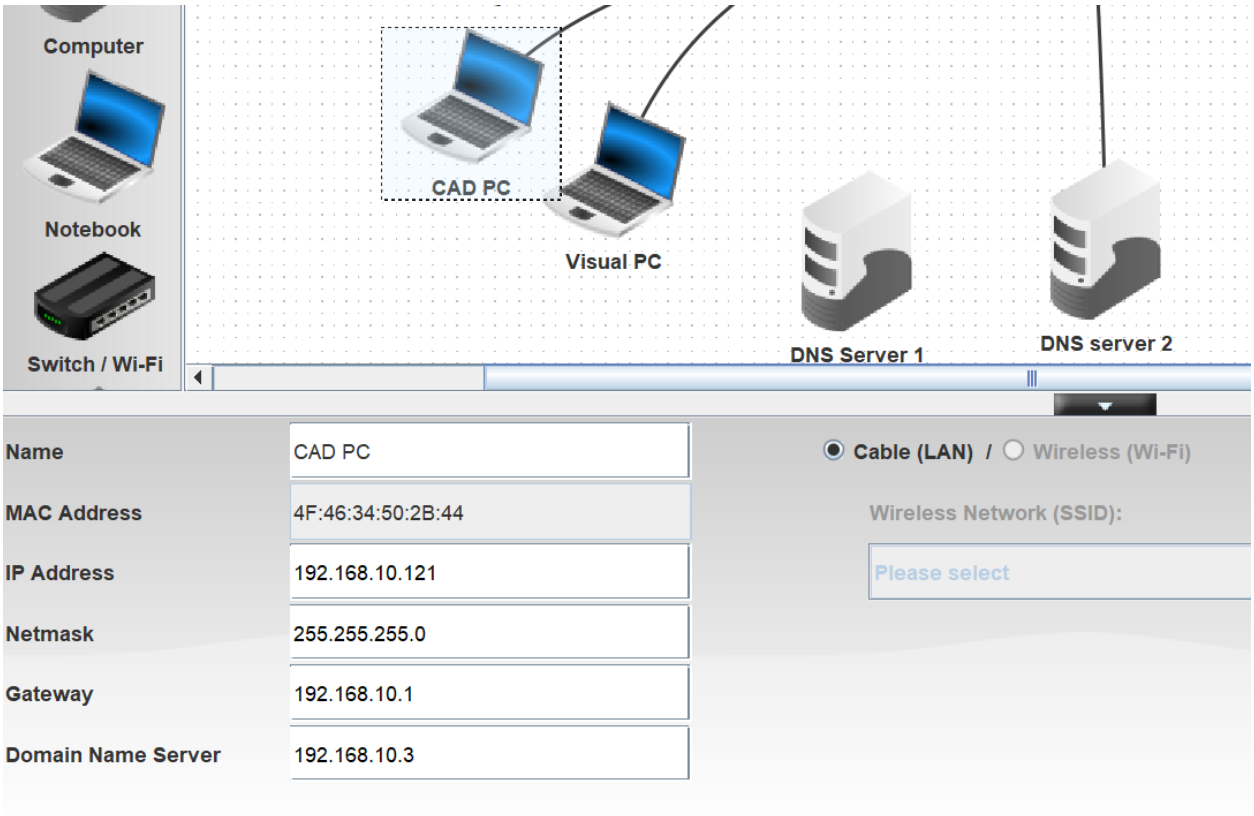


Figure 16: *The CAD PC's DNS Server was changed to (192.168.10.3 DNS) Server 2*

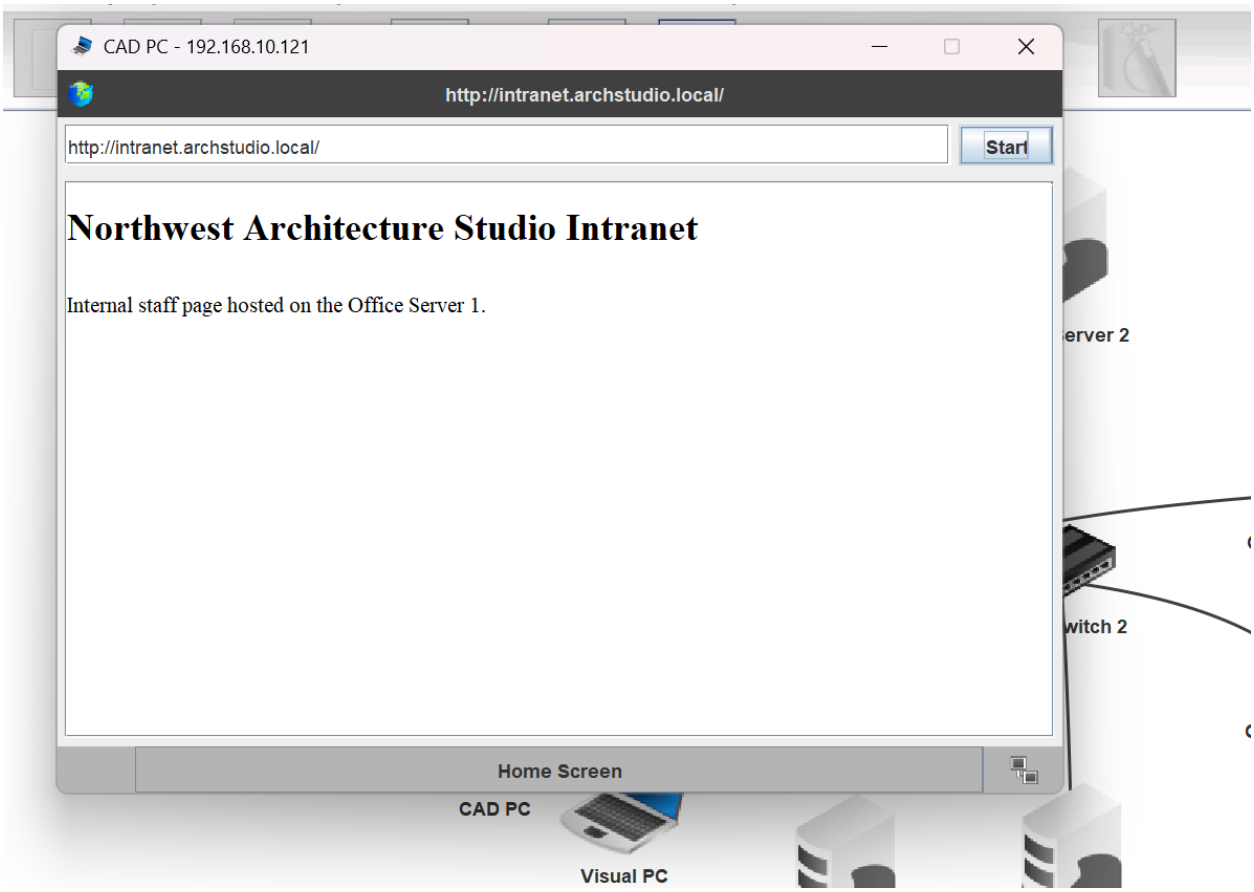


Figure 17: *The intranet opens by name again using DNS Server 2.*

LAN VS External Network:

The local area network (LAN) vs external network connects devices within northwest architecture studio including the admin pc project pc cad pc visual pc dns server and office server. These devices use the 192.168.10.0/24 network and communicate through the office switch. It allows employees to have access to the company's intranet and share information in the office. The LAN lets staff have shared access to files quickly without them needing an internet connection. This is very useful to cad designers who day in day out work with large files.

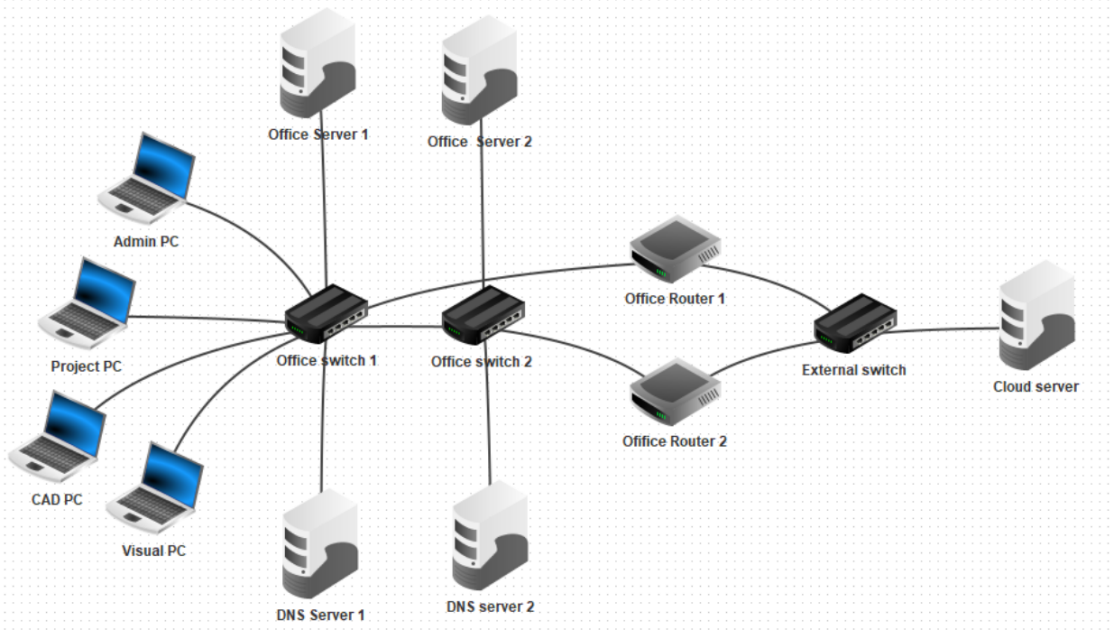
The external network represents the internet or cloud services. In the filius simulation, the cloud server uses Ip address 203.0.113.10 which connects through the isp switch. The office router connects both networks which allows the office computer to communicate with the external server. This allows employees to access external resources while remaining connected to the office network. The router uses ip addresses to direct the data between 2 separate networks

The main difference between them is that LAN is mostly used for internal communication whereas the external network provides access to the resources outside of the office. It separates the 2 networks and helps the company organise the network and it also provides a stable foundation for controlling the external access. The LAN is much faster for sharing local files because data doesn't need to travel outside of the office. So having separate networks makes it easier to manage traffic and to also apply security measures to protect company data.

Comparison

LAN	External Network
Internal office network	Outside of cloud network
192.168.10.0/24	203.0.0113.0/24
Uses office switch	Uses isp switch
Connects the office pcs and the servers	Connects cloud server

This is the completed filius network showing the internal office pcs switches routers dns severs, servers and external cloud servers.


Office Server 1
Office Server 2
Admin PC
Project PC
Office switch 1
Office switch 2
Office Router 1
External switch
Cloud server
Ofifice Router 2
CAD PC
Visual PC
DNS Server 1
DNS server 2

Team Contribution:

Key

Responsible = Main person who completed the task.

Consulted = Contributed feedback, ideas, checking, or support.

Task	Jimmy	Faris	Seerat	Aiman
Introduction	Consulted	Consulted	Consulted	Responsible
Objectives	Consulted	Consulted	Consulted	Responsible
Network Topology	Responsible	Consulted	Responsible	Consulted
IP Addressing Table	Consulted	Consulted	Responsible	Consulted
DNS Configuration	Consulted	Consulted	Responsible	Consulted
Client-Server Architecture	Consulted	Consulted	Responsible	Consulted
Protocol Explanations	Responsible	Consulted	Consulted	Responsible
Network Build (Filius Configuration)	Responsible	Consulted	Consulted	Consulted
Testing Table	Consulted	Responsible	Consulted	Responsible
Testing Screenshots/Evidence	Responsible	Consulted	Consulted	Responsible
Troubleshooting Log	Consulted	Consulted	Consulted	Responsible
Redundancy Section	Responsible	Consulted	Consulted	Consulted
LAN vs External Network	Consulted	Responsible	Consulted	Consulted
Report Formatting	Consulted	Consulted	Consulted	Responsible
Final Review	Consulted	Consulted	Responsible	Responsible

Conclusion:

This project successfully demonstrated the fundamental principles of computer networking through the design and implementation of a working network model in Filius. The completed network allowed devices on the office LAN to communicate with both internal and external services using switches, routers, DNS services, and web servers.

Testing confirmed successful communication between all devices and demonstrated the correct operation of TCP/IP, DNS, HTTP, ICMP, and gateway routing. Both internal and external web services were successfully accessed using IP addresses and domain names, demonstrating the client-server architecture used within modern networks.

The project also highlighted the importance of network resilience through the implementation of redundancy. By introducing backup servers, DNS services, and networking equipment, the simulated environment demonstrated how organizations can reduce service interruptions when failures occur.

Overall, the project achieved its objectives and provided practical experience in network design, configuration, testing, and troubleshooting while reinforcing key networking concepts.